

MSBA Job Matching - Final Technical Summary

Problem and Production Objective

The system supports a graduate business career advisor who must decide which opportunities to forward, hold, or investigate for MSBA international students. It accelerates first-pass review while keeping role fit, incomplete evidence, and work authorization as separate decisions.

Production Architecture

Stage	Production behavior
Route	Distinguish search requests from job postings
Retrieve	Enforce constraints, then rank and dedupe 80,000 postings
Classify	Use a 3.8 MB word+character TF-IDF SVM artifact
Govern	Calibrate uncertainty; override missing fields/restrictions
Respond	Return cited evidence and advisor action

Measured Improvement

Model	Training rows	Macro F1	Role
M2 baseline	80,000	0.791	Comparison
M3 PEFT LoRA	2,000	0.873	Adaptation experiment
Final TF-IDF SVM	77,135 deduped	0.984	Production decision layer

On the same 4,000-row subset, macro F1 improves by +0.192 over M2. Novel-text macro F1 is 0.979; unseen-company macro F1 is 0.980; newest-20% macro F1 is 0.976. The selective policy audit retains 91.6% coverage at 99.47% accuracy.

Why This Model

A training-size curve shows 0.898 macro F1 at 2,000 rows, 0.944 at 8,000, 0.965 at 20,000, and 0.984 at 80,000. The final linear model is more accurate than the current LoRA experiment, trains in under one minute, and is straightforward to reproduce. A larger transformer is not justified until human labels exist.

Real Runs and Failure Handling

Case	Model	System	Decision support
entry level analytics search	N/A (search)	N/A (search)	Advisor review before forwarding
senior engineering posting	low_fit	low_fit	Investigate - advisor review required
authorization restriction posting	high_fit	high_fit	Hold - explicit authorization restriction requires advisor verification
thin metadata posting	medium_fit	unclear	Investigate - advisor review required

Governance and Residual Risk

The casebook returns evidence for all four workflows, forces review for one explicit restriction, and contains zero unsupported authorization claims. Exact duplicates are removed; novel-text, unseen-company, and time-based slices are reported. Across 36 audited search results there are zero hard-constraint violations and zero duplicate company-title pairs. The full-source scan identifies 230 authorization candidates. Weak labels remain a limitation, so advisor fields are never auto-filled.

Business Recommendation

Use the included 1,000-row advisor queue to run a controlled Career Center review: 800 low-margin role-fit cases and 200 authorization candidates. Measure high-fit precision, minutes saved per posting, override rate, and authorization-evidence accuracy while keeping the system as decision support. The queue and annotation guide make that review reproducible without fabricating human labels.

Production Assets

The repository contains reproducible training, random/time/selective evaluation, hard-constraint search audit, full-source authorization audit, annotation queue, calibrated review policy, individual and batch CLIs, tests, model artifact, notebooks, governance documentation, slides, and demonstration workflows.